

FEMTIC-DABIC User Manual

A FEMTIC-derived program for 3-D MT inversion and model appraisal
ABIC, OCCAM, nonlinear L-curve, Lp regularization, model resolution, and linear uncertainty

Program version: v2.7.0.

FEMTIC-DABIC is a general-purpose 3-D magnetotelluric (MT) inversion program derived from FEMTIC. Its principal inversion method is a data-space variant of Akaike's Bayesian Information Criterion (D-DABIC). The program also supports OCCAM and nonlinear cubic-spline L-curve inversion.

The primary regularization types are L2, L1-style, and L0-style. Difference and Laplacian roughness operators, distortion correction, reference-model constraints, inversion model-resolution appraisal, and linear-uncertainty appraisal based on covariance diagonals are also available.

This manual contains the minimum information needed to understand the source layout and to build, configure, and run FEMTIC-DABIC.

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1. Program Overview

Relationship To FEMTIC

FEMTIC-DABIC is a maintained derivative of Yoshiya Usui's FEMTIC, not an independent rewrite. FEMTIC provides the mesh, observed-data, forward-modeling, Gauss-Newton inversion, sparse linear algebra, solver, and native I/O architecture.

Upstream FEMTIC:

<https://github.com/yoshiya-usui/femtic>

The FEMTIC-DABIC branch adds D-DABIC/ABIC, OCCAM, L-curve, Lp regularization, diagnostic, and appraisal functions.

Source Layout

Directory	Main responsibility
src/core	shared constants and program-level state
src/control	control.dat parsing and inversion dispatch
src/data	MT/VTF and other observed-data classes
src/mesh	finite-element mesh handling
src/forward	3-D electromagnetic forward calculation
src/inversion	model update and data-space/model-space inversion
src/model	resistivity blocks, roughening, and model output
src/linalg, src/solver	sparse matrices and solver interfaces
src/io	shared input/output helpers

Main Capabilities

Function	Maintained options
Main inversion methods	exact/inexact ABIC, exact/inexact OCCAM, nonlinear cubic-spline L-curve
Regularization types	L2, L1-style IRWLS, and L0-style IRWLS
Roughness operator	Difference filter or Laplacian L2
Additional modes	fixed alpha, linear L-curve, and data-fit cooling
Optional constraints	reference model / minimum norm, galvanic distortion
Inversion appraisal	model-resolution and linear-uncertainty (covariance) diagonal estimates
Parallel execution	MPI between processes and OpenMP/MKL within each process

License

FEMTIC-DABIC is distributed under the MIT License.

Original FEMTIC source:
Copyright (c) 2021 Yoshiya Usui

FEMTIC-DABIC modifications:
Copyright (c) 2025-2026 Han Song

Keep the original FEMTIC attribution in derived source files. MPI, Intel oneAPI/MKL, and other external dependencies retain their own licenses.

2. Build And Run

Build Environment

The maintained build targets Linux or WSL with Intel oneAPI MPI, OpenMP, MKL ILP64/PARDISO, GNU Make, and a compatible C++ compiler.

```
source /opt/intel/oneapi/setvars.sh
cd /path/to/FEMTIC-DABIC/src
make check-env
make -j2
```

The executable is written to:

```
src/femtic-dablc
```

Useful commands:

```
make print-config
make debug
make clean
```

MKLROOT defaults to /opt/intel/oneapi/mkl/latest and can be overridden on the make command line.

Run Directory

FEMTIC-DABIC reads and writes the current working directory. Keep each run in a separate directory containing at least:

```
control.dat
mesh.dat
observe.dat
resistivity_block_iter0.dat
```

Launch from that directory:

```
cd /path/to/run
export OMP_NUM_THREADS=2
export MKL_NUM_THREADS=2
mpirun -np 2 /path/to/FEMTIC-DABIC/src/femtic-dablc
```

Set NUM_THREADS in control.dat consistently with the OpenMP/MKL allocation. Choose MPI ranks, threads, memory, and wall time according to the number of frequencies and the mesh size.

3. Input Files

File	Purpose	Required
control.dat	inversion, solver, output, and stopping controls	yes
mesh.dat	FEMTIC 3-D finite-element mesh	yes
observe.dat	stations, frequencies, observed responses, and standard deviations	yes
resistivity_block_iterN.dat	initial resistivity blocks for starting iteration N	yes
Referencemodel.dat	reference model for minimum-norm/reference stabilization	only when enabled
distortion_iterN.dat	existing distortion state	only when read by the selected distortion setup

The initial model index must match the first value of `ITERATION`. Mesh, observations, model blocks, coordinates, units, station ownership, and frequency ordering must belong to the same case.

For raw `MT` and `VTF` data, each real or imaginary scalar is active only when its own standard deviation is positive. A standard deviation ≤ 0 masks that scalar from the residual, sensitivity matrix, RMS, and model update. This scalar masking rule must not be assumed for other observation types without checking the detailed input contract.

4. Core control.dat Settings

Keywords and values are whitespace-separated. Place settings before `END` and normally define each block once. Keyword order matters for method-specific options.

Keyword	Purpose
INV_METHOD	selects ABIC, OCCAM, L-curve, or an additional inversion mode
DATA_SPACE_METHOD	selects the data-space inversion formulation
TYPE_OF_TRADE_OFF_PARAMETER	supplies the method-specific numeric type required by the parser
TRADE_OFF_PARAMETER	supplies alpha, target RMS, or candidate values
ABIC_SEARCH_MODE	selects EXACT or INEXACT for ABIC only
DIFF_FILTER	enables Difference-filter roughness and selects its norm
DISTORTION	disables or selects distortion estimation
LEVENBERG_MARQUARDT	optionally adds non-negative LM damping to the model normal matrix
ITERATION	gives initial and maximum iteration indices
NUM_THREADS	records the thread allocation used by the run
FWD_SOLVER	selects the forward linear solver
MEM_LIMIT	sets the program memory limit
RESISTIVITY_BOUNDS	enables or disables model bounds
APPRAISAL_MODE	optionally requests model-resolution and linear-uncertainty/covariance diagonals
END	terminates control parsing

Omit `LEVENBERG_MARQUARDT` when LM damping is not intended. Writing the keyword with a zero value still activates its internal flag.

Inversion Methods

Method	INV_METHOD	TYPE_OF_TRADE_OFF_PARAMETER	Payload
Fixed alpha	1	0	alpha
ABIC	2	1	initial_alpha target_rms
OCCAM	3	2	initial_alpha target_rms
Linear cubic-spline L-curve	4	3	candidate-alpha definition
Nonlinear cubic-spline L-curve	5	4	first-iteration start alpha
Data-fit-bracketed cooling	6	5	initial_alpha target_rms

Do not use older OCCAM/L-curve method numbering with v2.7.0.

Data-Space ABIC Formulation

D-DABIC evaluates candidate models in data space and selects the alpha that minimizes ABIC. For each candidate alpha, the program forms a Gauss-Newton model update and evaluates the objective

$$\text{Phi}(\text{alpha}) = \text{Phi}_d + \text{alpha}^2 \text{Phi}_m + \text{alpha}^2 \text{gamma}^2 \text{Phi}_{\text{ref}}$$

where Phi_d is the weighted data misfit, Phi_m is the active Difference or Laplacian roughness, and gamma is the reference-model weight. The symbol beta is reserved for distortion-correction weights. The reference-model term is included only when that option is active. The ABIC scalar combines this objective with the data-space determinant term and the number of active data. The selected alpha therefore balances data fitting and model regularization statistically rather than by a prescribed alpha or target RMS alone.

One ABIC trial performs the data-space model solve, updates the candidate model and optional distortion parameters, calculates a full forward response, and then records RMS, roughness, model norm, model-update ratio, objective, and ABIC. It does not recalculate the sensitivity matrix for every alpha trial; the current iteration sensitivity is reused during the search.

The payload

```
TRADE_OFF_PARAMETER
10 0.01
```

contains `initial_alpha target_rms`. The first value starts the alpha search. The second is used by the accepted-update and stopping logic; it does not replace ABIC minimization as the alpha-selection criterion.

After selection, the program restores the selected trial model, distortion state, forward response, and residual. At the start of the next inversion iteration, the normal forward and sensitivity calculations are still performed.

ABIC supports Difference-filter L2/L1/L0-style regularization and Laplacian L2 roughness. Reference-model and distortion terms are included when their corresponding controls are enabled.

ABIC Exact And Inexact Search

Exact ABIC is the default when `ABIC_SEARCH_MODE` is omitted. It can also be selected explicitly:

```
INV_METHOD
2
ABIC_SEARCH_MODE
EXACT
```

Exact ABIC first brackets the minimum in $\log_{10}(\alpha)$ with an initial span of 0.25 and then refines it by Brent minimization.

The maintained efficient ABIC workflow is:

```
INV_METHOD
2
ABIC_SEARCH_MODE
INEXACT
```

Inexact ABIC selects alpha from the bracket samples without Brent refinement. It uses a $\log_{10}(\alpha)$ span of 0.5 on the first update and 0.2 on later updates. If the accepted full-step model is rejected, it retries the same alpha at the reduced step before starting another search; it does not fall back to exact refinement.

For one transition release, legacy `RUN_INEXACT_LINE_SEARCH 1 2` maps to `ABIC_SEARCH_MODE INEXACT`. Legacy modes 1 0, 1 1, 1 3, and 1 4 are rejected.

OCCAM Exact And Inexact Search

OCCAM uses `INV_METHOD 3`. Without an OCCAM inexact keyword, the program uses the exact Brent/legacy OCCAM search:

```
INV_METHOD
3
DATA_SPACE_METHOD
2
TYPE_OF_TRADE_OFF_PARAMETER
2
TRADE_OFF_PARAMETER
10 0.01
```

To use the inexact Phase-I search, insert:

```
RUN_INEXACT_OCCAM_LINE_SEARCH
1
```

The inexact option accepts the bracketed Phase-I RMS minimum without Brent refinement. The remaining OCCAM target-RMS search follows the common OCCAM path.

Nonlinear L-Curve

Nonlinear cubic-spline L-curve evaluates actual forward responses, removes the low-alpha rising tail, and selects the retained-curve corner. It starts from one alpha:

```
TRADE_OFF_PARAMETER
100
```

Additional Modes

Fixed-alpha inversion applies a prescribed alpha. Linear L-curve uses a fixed candidate-alpha list and supports log/linear and root/squared coordinate forms. Data-fit cooling adjusts alpha from actual RMS decrease:

```
ALPHA_COOLING
0.20 0.05 0.9 0.1
```

The four values are the initial required RMS decrease, later cooling trigger, alpha multiplier, and minimum alpha. Cooling is maintained for Difference L2 and Laplacian L2.

5. Regularization Types And Optional Functions

L2, L1, And L0

Include `DIFF_FILTER` to use Difference-filter regularization. Its first value selects the regularization type:

Value	Type	Interpretation
2	L2	quadratic smoothing that distributes model gradients
1	L1-style IRWLS	iterative reweighting that reduces the penalty on persistent gradients
0	L0-style IRWLS	minimum-gradient-support proxy that favors compact nonzero gradients

L1 and L0 are implemented through iteratively reweighted least squares; they are numerical approximations rather than direct nonsmooth solves.

Example Difference-filter L2 block:

```
DIFF_FILTER
2
0.01 2.0
1
1
```

For L1, the following values define the gradient limits, maximum IRWLS iterations, and convergence ratio. For L0, they define the small value, maximum IRWLS iterations, and convergence ratio. A practical initial L0 small value is 0.01, but it is not a universal geological optimum.

Difference And Laplacian Operators

Difference regularization penalizes neighboring-model differences. Omit the complete `DIFF_FILTER` block to use the Laplacian L2 roughness operator.

Levenberg-Marquardt Damping

LM damping is an optional stabilization of the model update. The current implementation replaces the model normal matrix by

```
R^T R + lambda_LM I
```

before the data-space solve. A positive `lambda_LM` improves conditioning and usually suppresses large model updates, but excessive damping can slow data fitting. Enable it with:

```
LEVENBERG_MARQUARDT
<lambda_LM>
```

The damping value must be non-negative. When `SMALL_VALUE` is also active, the two diagonal additions are summed. LM damping changes the linearized model update; `STEP_LENGTH` instead scales or retries the update after it has been solved. The run summary reports either Gauss-Newton Or Levenberg-Marquardt damping, damping = <value>.

To disable LM damping, omit the complete keyword block. Do not use a zero value as an off switch because the keyword itself enables the LM branch.

Reference Model And Distortion

Reference-model/minimum-norm stabilization uses:

```
TYPE_OF_REFERENCE
0
WEIGHT_OF_REFERENCE
<weight>
NORM_OF_MINIMUMNORM
2 0.01 2.0
```

The three weights are distinct: `alpha` controls model roughness, `beta` controls distortion correction, and `gamma` is read from `WEIGHT_OF_REFERENCE`. For fixed-alpha inversion, the corresponding objective can be written as

```
Phi = Phi_d + alpha^2 Phi_m + beta^2 Phi_dist + gamma^2 Phi_ref
```

Use the following core block for fixed-alpha inversion with a fixed reference model:

```

INV_METHOD
1
DATA_SPACE_METHOD
2
TYPE_OF_TRADE_OFF_PARAMETER
0
TRADE_OFF_PARAMETER
10.0
DISTORTION
0
WEIGHT_OF_DISTORTION
TYPE_OF_REFERENCE
0
WEIGHT_OF_REFERENCE
1.0
NORM_OF_MINIMUMNORM
2 0.01 2.0

```

Place the reference model in the run directory as `Referencemodel.dat`. It must use the same element/block mapping as the active resistivity model and contain positive resistivity values. `TYPE_OF_REFERENCE 0` keeps this model fixed throughout the inversion, while `WEIGHT_OF_REFERENCE` sets its stabilization weight. Keep the mesh, solver, output, iteration, convergence, step-length, and roughness-operator blocks from the complete example below.

Distortion modes are:

DISTORTION	Meaning
0	disabled
1	distortion-matrix difference
2	gains and rotations
3	gains only

Inversion Resolution And Linear-Uncertainty Appraisal

The module uses run-local sensitivity matrices and the active roughening operator. The resolution diagonal measures each parameter's linearized resolving strength; the covariance diagonal measures linear inversion uncertainty under the active data weighting and regularization. These are diagonal estimates, not complete matrices or nonlinear posterior intervals.

Omit `APPRAISAL_MODE` to disable appraisal. Otherwise:

APPRAISAL_MODE	Output
0	model-resolution and linear-uncertainty/covariance diagonals
1	model-resolution diagonal only
2	linear-uncertainty/covariance diagonal only

```

APPRAISAL_MODE
0
APPRAISAL_RANDOM_VECTORS
8192

```

The program reads run-local sensitivity files and writes appraisal results under `appraisal/`.

6. Complete Exact ABIC Example

The following Difference-filter L2 control is a compact template. Mesh, solver, memory, bounds, output, and stopping values must be reviewed for the actual case.

```
INV_METHOD
2
ABIC_SEARCH_MODE
EXACT
DATA_SPACE_METHOD
2
TYPE_OF_TRADE_OFF_PARAMETER
1
TRADE_OFF_PARAMETER
10 0.01
DISTORTION
0
WEIGHT_OF_DISTORTION
DIFF_FILTER
2
0.01 2.0
1
1
MESH_TYPE
2
NUM_THREADS
2
FWD_SOLVER
0
MEM_LIMIT
3000
DIV_NUM_RHS_FWD
1
DIV_NUM_RHS_INV
1
ELEC_FIELD
0
OWNER_ELEMENT
-1
RESISTIVITY_BOUNDS
0
OUTPUT_PARAM
1
0
ITERATION
0 2
CONVERGE
1
ROUGH_MATRIX
0
DECREASE_THRESHOLD
0.1
STEP_LENGTH
1.0 0.1 1
1
0.5 1.5
RETRIAL
3
APP_PHS_OPTION
1
BOTTOM_RESISTIVITY
300.0
END
```

To use Laplacian roughness, remove the entire DIFF_FILTER block. To run inexact ABIC, replace EXACT with INEXACT.

7. Output Files

Common native outputs include:

```
FEMTIC-DABIC_<rank>.log  
FEMTIC-DABIC.cnv  
FEMTIC-DABIC_iter<iter>.cnv  
result_<rank>_iter<iter>.csv  
resistivity_block_iter<iter>.dat  
Resistivity.iter<iter>  
Mesh.geo  
BlockIDs  
sensMatFreq*
```

FEMTIC-DABIC.cnv contains accepted-iteration quantities such as RMS, selected alpha, model roughness, model update, and method-specific objective values. Iteration-specific CNV and diagnostic CSV files record trial behavior. The rank logs contain parser, solver, forward, and inversion messages. Model files store the accepted resistivity blocks for each written iteration.

8. Important Boundaries

- The selected-trial cache restores MT/VTF responses and residuals only; the next inversion iteration still performs its normal forward and sensitivity calculations.
- Hessian-family diagonal appraisal is not part of the active v2.7.0 scope.
- L-curve with distortion or reference-model stabilization is outside the primary supported combinations described in this manual.
- Laplacian plus inexact OCCAM plus distortion should be treated as an advanced diagnostic combination.
- A control file from an older FEMTIC-DABIC version must be migrated rather than silently reused.

Additional parser and file-format details are available in:

```
maintenance/docs/control_keywords_3d_abic.md
maintenance/docs/INPUT_FORMAT.md
maintenance/docs/IO_CONTRACTS.md
maintenance/docs/VERSION_HISTORY.md
```